EE650 · BASICS OF MODERN CONTROL SYSTEMS · PROGRESS REVIEW 1

Fault-tolerant control of a quadrotor after complete rotor loss

Residual Koopman MPC, and certified limits on fault-detection delay

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RECAP

Two research questions

RQ1 · AS PROPOSED

Residual Koopman MPC

Can a Koopman (EDMDc) model of only the mismatch, built on the known spinning-equilibrium model of Mueller & D'Andrea, predict and control better than the physics model alone or a full-state Koopman model?

RQ2 · ADDED AFTER THE PROPOSAL

Certified detection-delay budget

A fault detector needs 30–130 ms to notice a dead rotor. What is the largest delay for which recovery is guaranteed, and how do the RQ1 controllers change it?

The link: each capture controller (LQR, analytical MPC, residual Koopman MPC) is compared by the detection delay it can tolerate.

THE PROBLEM

What happens when a rotor fails

Healthy flight
Normal controller, four rotors

→

Rotor fails
t = 0, thrust of rotor 4 lost

→

Undetected, τ
Healthy controller still flying a 3-rotor vehicle

→

Capture
Failure controller takes over

→

Relaxed hover
Steady spin ≈ 19 rad/s, axis tilted

Most recovery methods with a guarantee assume τ = 0.

Stability with a fixed delay is shown for a linear model; a certified maximum delay under thrust limits was not found.

Simha et al. 2017; Al Lawati & Akhtar 2026; Başak & Prempain 2015; Strack van Schijndel et al. 2021

Simulator and vehicle

- 6-DOF rigid body, quaternion attitude, rotor gyroscopic terms
- Per-rotor thrust 0.2–3.8 N; a motor can also be stopped
- First-order motor lag, 15 ms
- Reduced attitude $n = R^T e_3$: the "up" direction seen from the body
- RK4 integration; controller at 1–5 ms (zero-order hold)
- Truth plants for learning add thrust loss at high spin, inertia error and a different motor lag

Parameter	Value
Mass	0.50 kg
Arm length	0.17 m
Inertia Ixx, Izz	3.2, 5.5×10^{-3} kg m²
Thrust coefficient κf	6.41×10^{-6} N s²
Torque ratio κτ	0.0169 m
Yaw drag γ	2.75×10^{-3} N m s

All values from Mueller & D'Andrea, ICRA 2014, Sec. V-A

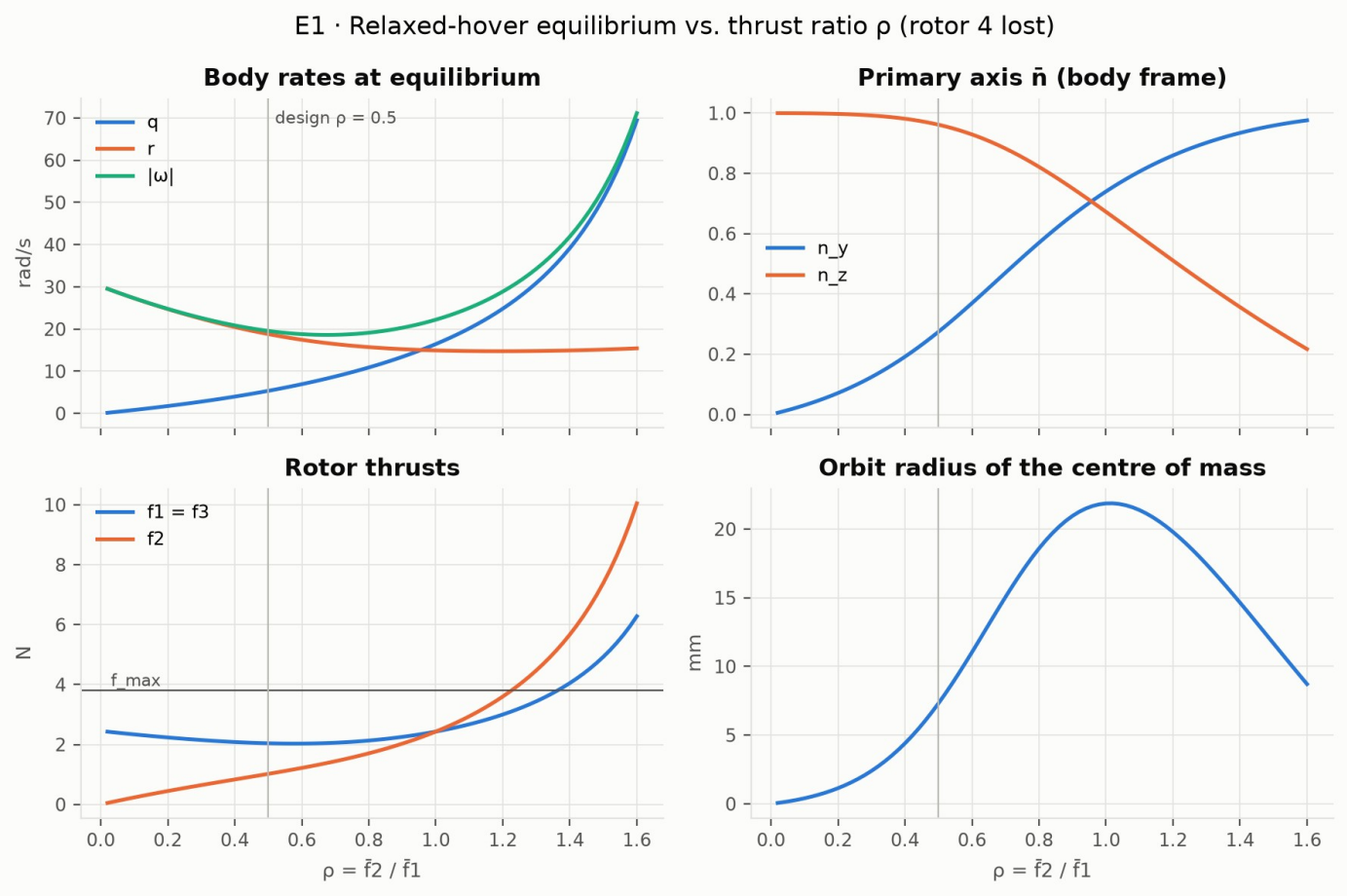
FOUNDATION

The model matches three published papers

Source	What we compared	Result
ICRA 2014	Two-rotor-loss equilibrium	All published digits (≤ 0.14%)
ICRA 2014	Three-rotor-loss equilibrium	All published digits (≤ 0.8%)
ICRA 2014	One-rotor-loss equilibrium	f, r, n_z ≤ 0.5%; pitch rate −6% *
ICRA 2014	ā and controllability loss (eqs. 28, 39)	Exact
IJRR 2016	Five failure cases, own implementation	0–4%; ≤ 0.7% within rounding
Sun et al. 2020	Bebop 2, two-rotor loss (eq. 60)	Exact
Flight data	Time to spin up to 10 rad/s	**0.80 s vs 0.62 s: model 30% slower**

* The paper's own published values do not satisfy its eq. (12); its other two cases match exactly. **45 automated tests** also check conservation laws, closed-form solutions and these published numbers.

The spinning equilibrium



Rotor 4 lost, design ratio $\rho = f_2/f_1 = 0.5$:

f = (2.04, 1.02, 2.04) N
spin 18.8 rad/s, axis tilted 16°

Thrust, spin rate and tilt match the paper within 0.5%.

The ρ sweep reproduces the paper's Fig. 3 (minimum spin 18.6 vs 19.0 rad/s).

Linear model: controllable, but what can we observe?

7 states (p, q, r, n_x, n_y, z, v_z), inputs f_1, f_2, f_3. Linear and time-invariant in body axes.

Open-loop modes: ±10.1j, ±19.6j, −0.51 and an altitude double integrator.

Controllable: rank 7 of 7; every mode passes the PBH test.

Sensors	Rank	Missing
Gyro	3 / 7	tilt, altitude
Gyro + tilt estimate	5 / 7	altitude
Gyro + barometer	7 / 7	none (weak)
Gyro + tilt + barometer	7 / 7	none

The equilibrium is tilted, so vertical acceleration depends on tilt. A barometer alone then reveals the tilt, but about 2,700× more weakly than a tilt sensor (observability Gramian).

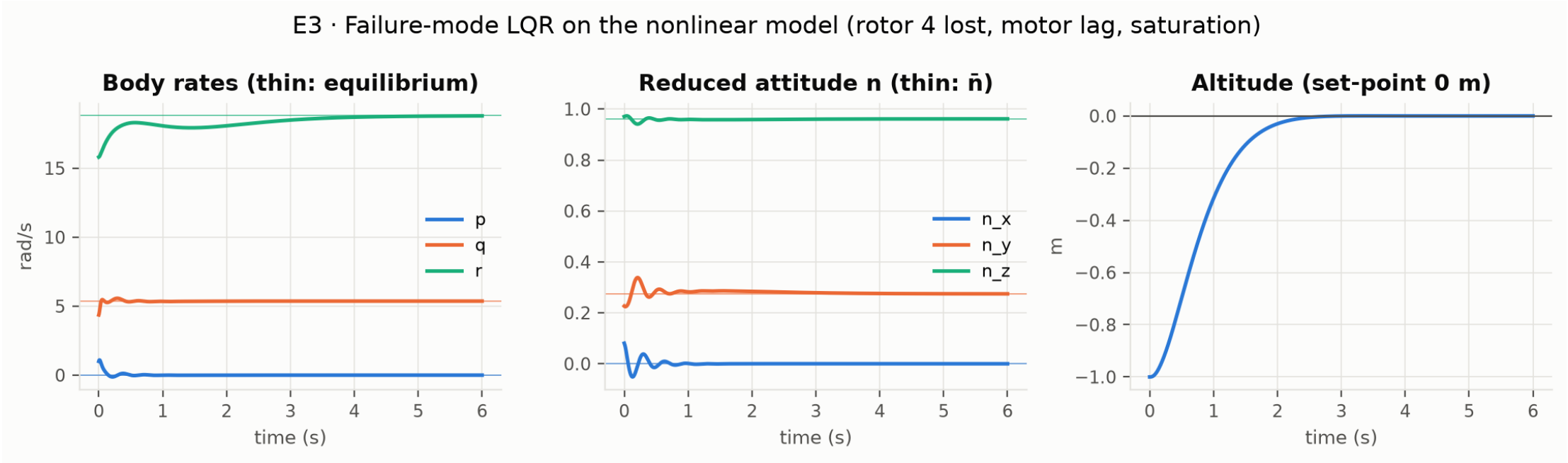
FOUNDATION

LQR baseline at the spinning equilibrium

18 ms
input-delay margin (healthy LQR: 40 ms)

ζ ≈ 0.19
damping of the slowest oscillation

1–4 s
to settle tilt, altitude and spin



Weights from Mueller & D'Andrea, plus motor states and altitude

RESEARCH QUESTION 1 · AS PROPOSED

Residual Koopman models of the failed vehicle

Learn only what the physics model gets wrong, then use it inside a convex MPC.

Three predictors, same data

Analytical

Mueller & D'Andrea linear model about the spinning equilibrium. No data.

Full-state EDMDc

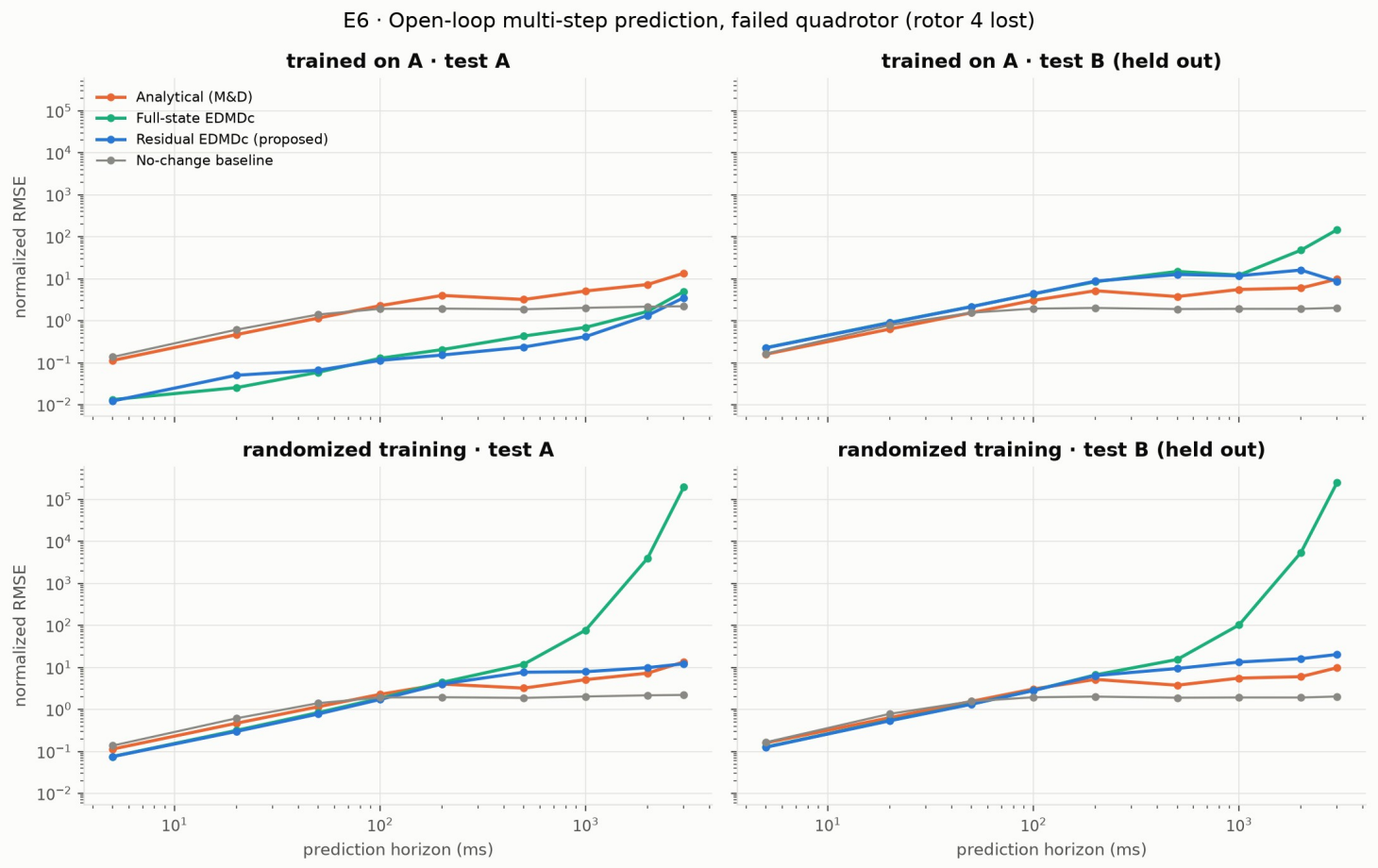
Learns all the dynamics from data: 55 lifted states (quadratic dictionary).

Residual EDMDc (proposed)

Physics model plus a learned one-step correction e = x_true − x_model, clipped at d_max.

Data: 40 episodes × 4 s at 5 ms, from a truth plant with effects the model lacks (thrust loss at high spin, inertia error, slower motors). **Test:** 10 new episodes from the same plant (A) and 10 from a different one (B).

The residual model predicts best

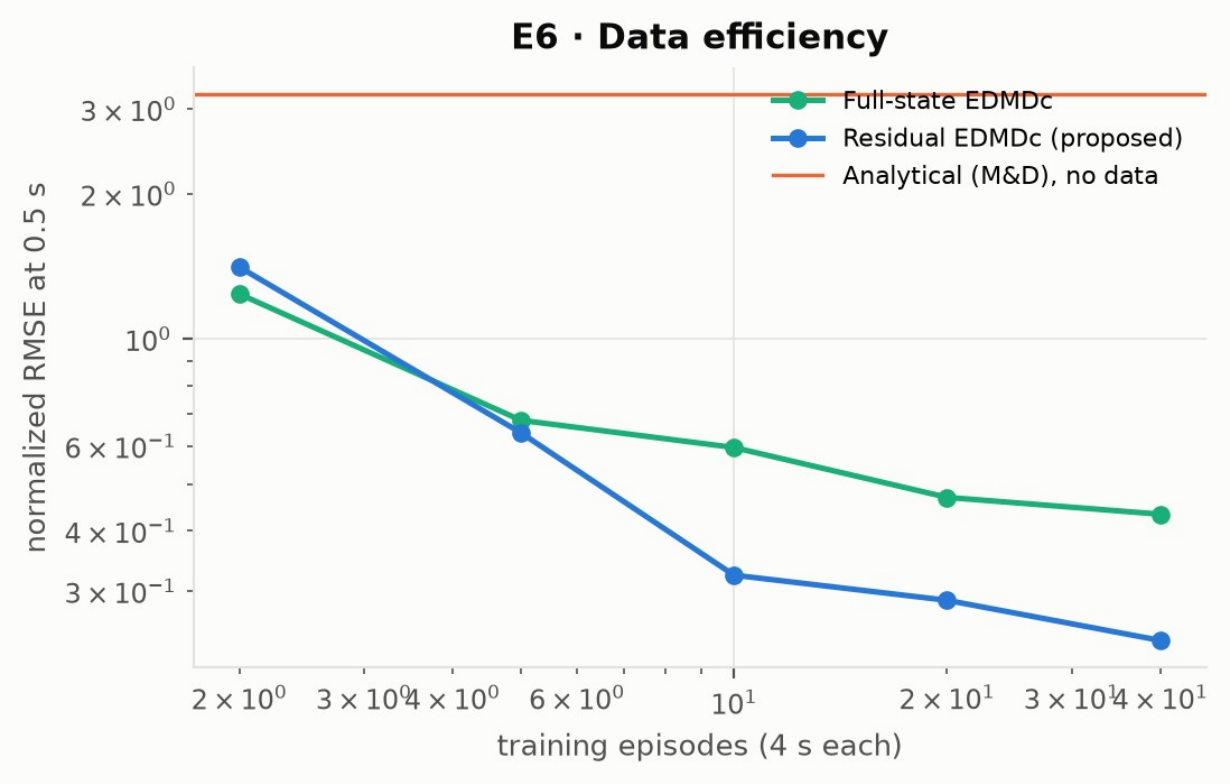


Error at a 1 s horizon, test family A (normalized RMSE):

Predictor	Error
Residual EDMDc	**0.42**
Full-state EDMDc	0.70
"No change" baseline	2.0
Analytical model	5.1

Full-state EDMDc has 4 of 55 eigenvalues outside the unit circle and falls behind "no change" after 2–3 s, as the proposal predicted.

Little data needed, no generalization yet



Data efficiency

With 10 episodes the residual model (0.32) beats the full-state model trained on 40 (0.43), at a 0.5 s horizon.

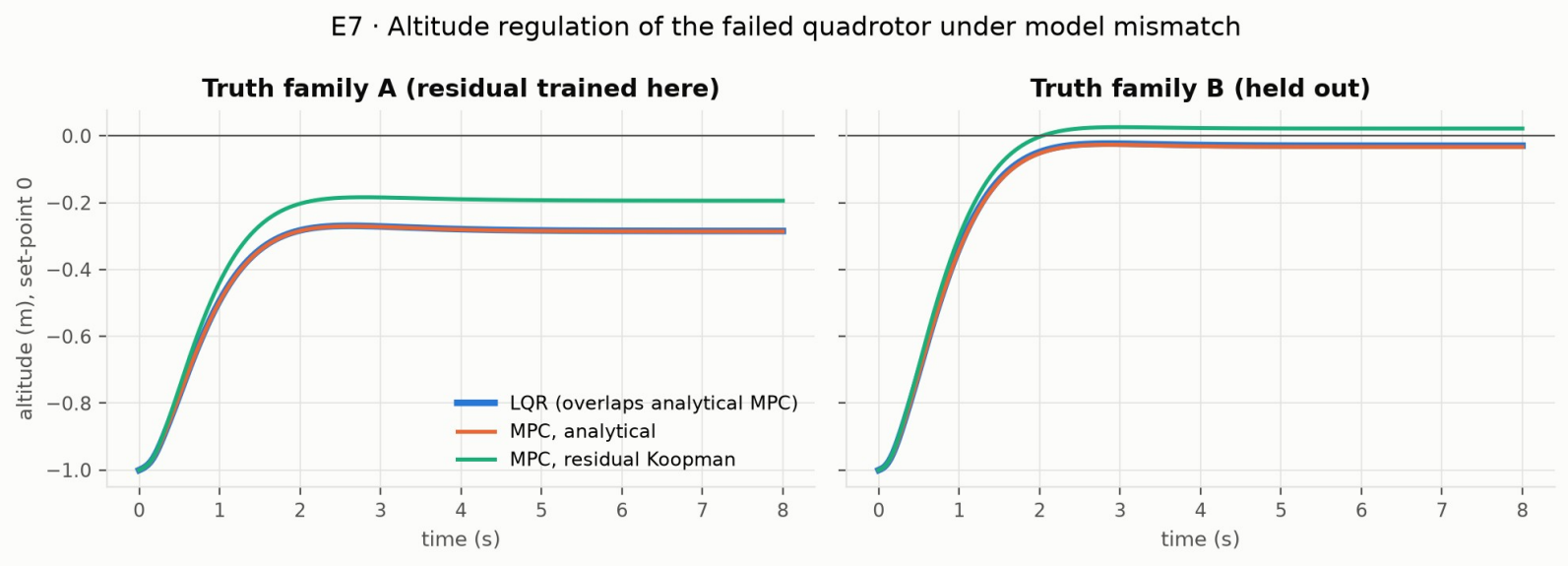
Generalization fails

On the unseen plant B, both learned models are worse than the physics model alone. Training on randomized plants makes them worse: the unknown parameters cannot be read from one state sample.

Next

Update the residual online with recursive least squares (it is linear in its parameters).

Residual MPC removes a third of the bias



−32%

steady altitude error on plant A (−0.29 → −0.19 m)

≈ 5 ms

per QP (horizon 20 × 5 ms, OSQP)

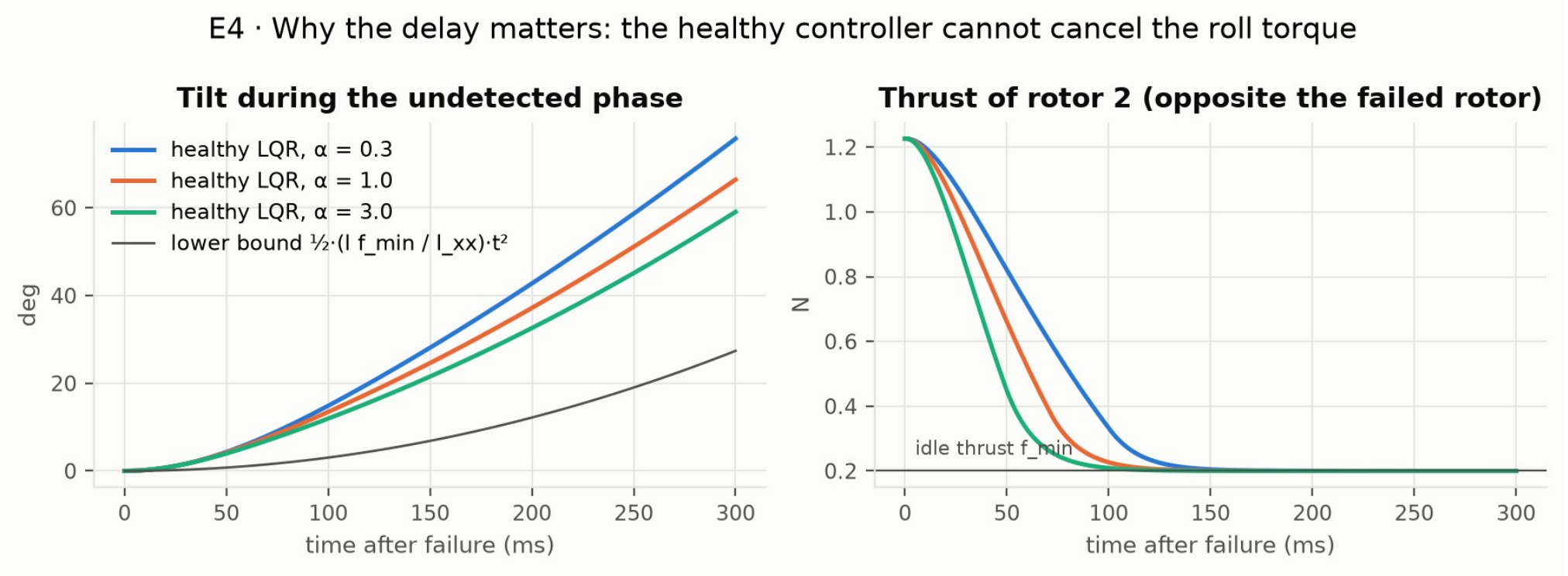
The residual enters the QP as a measured disturbance, so the problem stays convex (as in Todde et al. 2026, who report 35% in healthy flight). No harm on plant B. As the controller that catches the vehicle after a detection delay, neither MPC beats LQR yet.

RESEARCH QUESTION 2 · ADDED

How long can fault detection take?

Most recovery methods assume instant detection. Fast detectors take 30–130 ms.

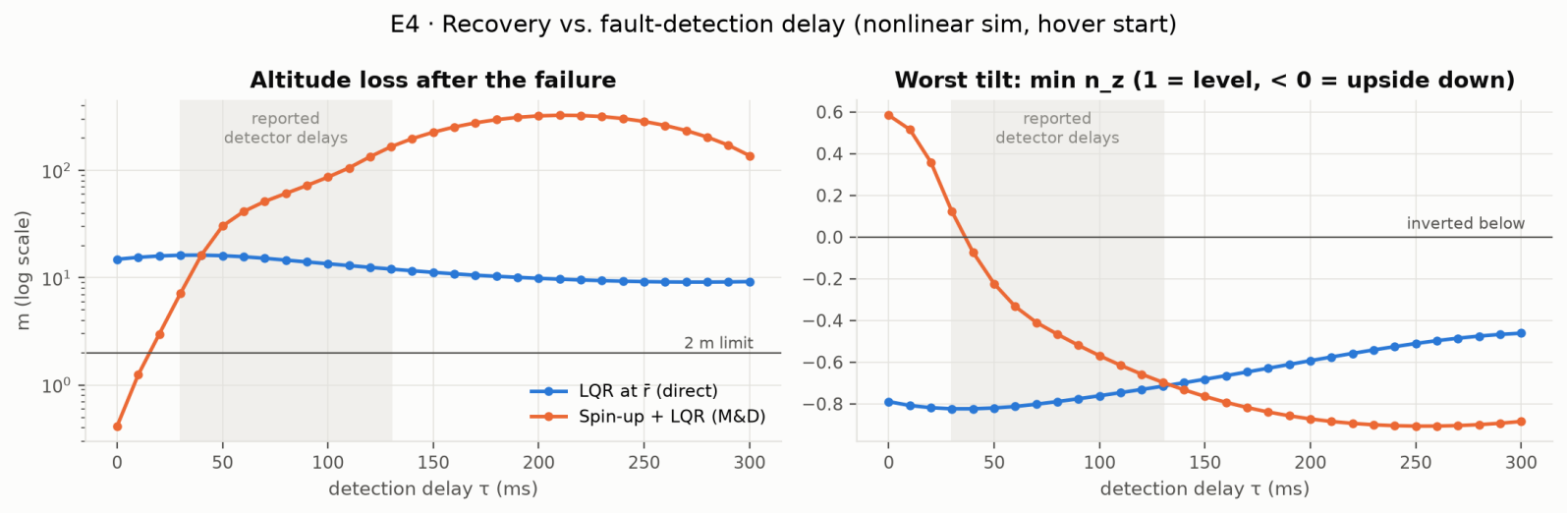
Before detection, nothing stops the roll



Rotor 2 cannot go below its idle thrust (0.2 N), so the roll acceleration is at least **l·f_min / I_xx = 10.6 rad/s²** for any healthy-controller tuning. Tilt reaches about 13° at 100 ms and 37° at 200 ms.

RQ2 · KEY RESULT

The published strategy tolerates about 10 ms

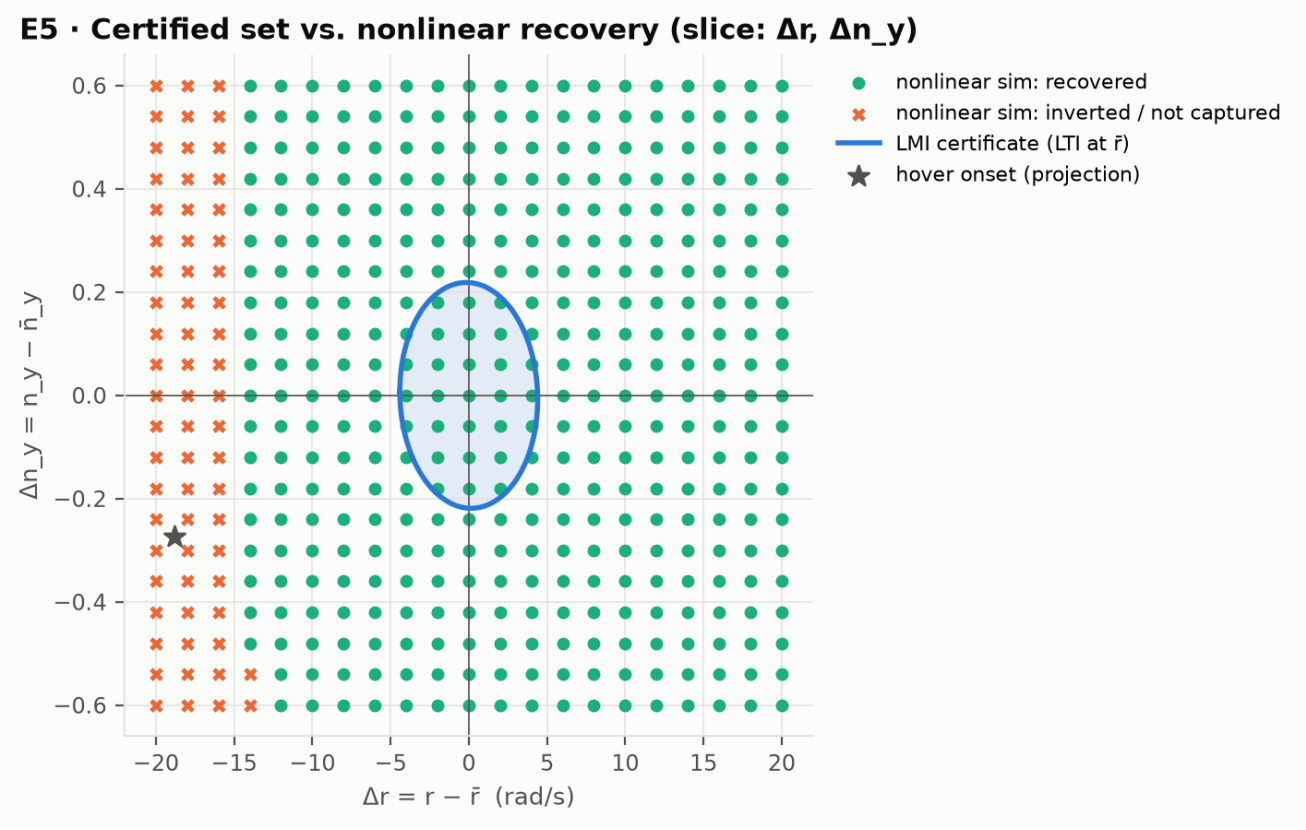


Safe recoveries, 20 random starts:

Delay	Safe
0 ms	20 / 20
10 ms	17 / 20
20 ms	6 / 20
30 ms	1 / 20
50 ms	**0 / 20**

Safe = attitude recovered, altitude loss under 2 m, never upside down (fixed before the runs). At τ = 0 the strategy loses 0.4 m, consistent with the paper's flight. With the looser "attitude recovered" criterion the tolerance is about 40 ms. Our model spins up more slowly than the real vehicle, so these numbers hold for the published model.

A first LMI certificate, and its limit



Saturation-aware invariant ellipsoid (Hu–Lin LMIs) for the LQR on the linear model at the spinning equilibrium.

22%
of the way to hover onset, certified

70%
recovered in simulation

The LQR fails only when the spin is near zero, exactly where the hover-onset state lies. Next: a certificate valid over the whole spin-up, treating the yaw rate r as a varying parameter (LPV).

NEXT STEPS

Plan for the remaining weeks

Weeks	RQ1 · Residual Koopman	RQ2 · Delay budget
1–2	MPC on the full-state Koopman model; integral-action MPC baseline	Read Stephan 2018 and Lanzon 2014; certificate over the spin-up (LPV in r)
3–4	Online residual adaptation (recursive least squares)	Reachable set of the undetected phase; first certified delay budget τ^*
5–6	Air-relative thrust in the truth plant; residual trained on spin-up data	Trade study incl. yaw-drag and thrust-model sensitivity; τ^* per controller
7–8	Ablations and write-up	Simulation check of the certified τ^*; final report

Main risk: a certificate over the whole spin-up may be infeasible or too conservative. Fallbacks: a parameter-dependent Lyapunov function, or a chain of nested ellipsoids.

SUMMARY

Where we stand

1 The simulator matches three published papers. Known gap: the real vehicle spins up about 30% faster.

2 RQ1: the residual Koopman model predicts best, needs less data and cuts the MPC altitude bias by a third, but does not yet generalize to an unseen vehicle.

3 RQ2: the published recovery tolerates about 10 ms of detection delay; detectors need 30–130 ms. Certifying the spin-up is next.

Questions?